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Increasing access to assistive technology and rehabilitation for people living with diabetes

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The number of people with diabetes around the world is rising, with prevalence increasing more rapidly in low- and middle-income countries.

Yet, many people with diabetes around the world do not have access to the essential care they need to manage their condition and help reduce the risk of complications, including blindness, kidney failure, heart attacks, stroke, and lower limb amputation.

Assistive technology and rehabilitation play an important role in addressing some of the complications associated with diabetes. For example, a person with diabetes can benefit from rehabilitation involving aerobic, resistance, flexibility, balance and walking exercises for improved sensation, blood flow, joint mobility, mobility and diabetes control in general. A person with a diabetic foot ulcer may require an assistive product such as a rigid removable boot, walking stick or therapeutic footwear, to promote ulcer healing, whilst maintaining mobility.



Photo credit: Louise Puli

The World Health Organization (WHO) reports a global shortage of trained health personnel to provide assistive technology and rehabilitation, estimating that access to assistive technology is as low as 3% in some countries, (1) and in certain low- and middle-income countries, there are fewer than 10 qualified rehabilitation practitioners per 1 million people. (2)

In response to this gap, WHO has developed [Training in Assistive Products](#) (TAP) and a [Basic Rehabilitation Package Clinical Resource](#) (BRP-CR). The aim is to equip

primary health personnel with the knowledge and skills to provide simple assistive products and basic rehabilitation interventions.

TAP includes training modules on some assistive products commonly needed by people living with diabetes-related complications, like walking aids, therapeutic footwear, magnifiers and toilet and shower chairs. The BRP-CR encompasses training modules covering physical exercise, muscle strengthening, joint and muscle flexibility, walking, and self-care. These components play a crucial role in controlling and managing diabetes. Additionally, the resource offers education and advice on environmental adaptations for individuals experiencing low vision due to diabetes.



Photo credit: Abena Tannor

In 2021 and 2022, WHO trialled the modules relevant to diabetes complications in Fiji and Ghana. Building on existing noncommunicable disease (NCD) and diabetes initiatives, WHO worked with the Fiji Ministry of Health and Medical Services and the Ghana Health Service to strengthen their existing primary health workforces to provide simple assistive products and basic rehabilitation for self-care and mobility.

Across the 2 countries, 53 (3) primary health personnel (4) were trained to provide simple assistive products and basic rehabilitation interventions. Supported by rehabilitation professionals serving as local mentors, primary health personnel who participated in the training reported:

"Since the training, every month I go to services to share training on assistive products and rehabilitation. This is something I never did before. I think the training has helped in our government hospital especially management and referral of diabetic foot wounds." Learner, Ghana.

Overall, the response to training from both personnel and mentors was positive and reinforced the need for scaling up assistive technology and rehabilitation services for people living with diabetes.

The initiatives have also provided an opportunity to expand efforts to increase access to assistive products and rehabilitation.

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There is a high need in both countries for assistive technology and rehabilitation services for people living with diabetes and its related complications. The

initiatives in Fiji and Ghana have showcased how these essential services can be successfully integrated into primary health care.

"I see the benefit, and there are many people in Ghana that need rehabilitation. With this training, I now understand how to provide some assistive products and rehabilitation." Learner, Ghana.

For additional information, please contact:

- WHO headquarters, Assistive Technology team, assistivetech@who.int
Learn more about [Training in Assistive Products \(TAP\)](#).
- WHO headquarters, Rehabilitation Programme, rehab@who.int
Learn more about the [Basic Rehabilitation Package Clinical Resource \(BRP-CR\)](#).

(1) World Health Organization (2023) [Assistive technology – challenges](#), accessed 12/10/23.

(2) Need to scale up rehabilitation. Background paper for the WHO Rehabilitation 2030 meeting. Geneva, Switzerland: World Health Organization; 2017, accessed 12/10/23.

(3) 34 primary health personnel in Fiji and 19 in Ghana.

(4) Nurses, community health nurses and physician's assistants.